

UNIT 4

Environmental Sanitation

Environmental sanitation refers to the control of environmental factors that can potentially affect human health. It includes management of waste, safe water supply, hygiene practices, vector control, food sanitation, and proper maintenance of buildings and public facilities.

1. Environmental Sanitation Methods for Hostels & Hotels



A. Water Supply & Drinking Water Safety

- Ensure **continuous, potable water supply**.
- Regular cleaning and disinfection of overhead tanks (once every 3–6 months).

- Provide **RO/UV water purification** in dining halls and rooms.
- Maintain residual chlorine at **0.2–0.5 mg/L** at consumer end.

B. Food Sanitation

- Kitchen floors, cutting boards, and food-contact surfaces must be **washed and disinfected regularly**.
- Proper food storage at safe temperatures:
 - Cold storage: $\leq 5^{\circ}\text{C}$
 - Hot holding: $\geq 60^{\circ}\text{C}$
- Avoid cross-contamination (separate raw and cooked food).
- Staff must use **aprons, gloves, head caps**, and undergo hygiene training.

C. Solid Waste Management

- Segregation at source:

Biodegradable → Non-biodegradable → Recyclables

- Food waste composting using:
 - **Vermicomposting**
 - **Mechanical composters**
 - **Biogas units**
- Regular collection by municipal authorities.

D. Liquid Waste / Sewage Management

- Hotels (>20 rooms) should ideally have **STP (Sewage Treatment Plant)**.
- Maintain grease traps in kitchens to avoid blockage.
- Treated wastewater can be reused for gardening or flushing.

E. Vector Control

- Weekly fogging (if required), sealing of gaps/holes, mesh windows, and covering of drains.
- Eliminate mosquito-breeding sites (stagnant water).

F. Building Sanitation

- Daily cleaning of rooms, corridors, washrooms using disinfectants.
- Adequate ventilation and lighting.
- Pest control services at regular intervals.

2. Environmental Sanitation for Hospitals



A. Water Sanitation

- Continuous supply of **potable and soft water**.
- Regular disinfection of tanks and pipelines.
- Water quality testing monthly (microbial & chemical).

B. Hospital Waste Management

Follow **BMWM Rules 2016 (amended 2018 & 2019)**:

Waste Type	Colour Bag	Treatment
Infectious waste	Yellow	Incineration/Deep burial
Contaminated plastics	Red	Autoclaving → Shredding
Glass, metallic waste	Blue	Disinfection → Recycling
Sharp waste	White (translucent)	Autoclave → Shred

C. Infection Control Practices

- Hand hygiene (WHO 7-step method).
- Sterilization of instruments (autoclaving, chemical sterilants).
- Cleaning schedules for wards, ICUs, OTs:

- High-touch surfaces cleaned thrice daily.
- Operation theatres disinfected after every surgery.
- Isolation rooms for infectious cases.

D. Sanitary Facilities

- Adequate toilets with disinfectant cleaning every 2–3 hours.
- Proper ventilation in wards and OPDs.

E. Vector & Rodent Control

- Regular fumigation, sealing of cracks.
- Eliminate stagnant water.

3. Sanitation for Swimming Pools & Public Bathing Places



A. Water Quality Management

- Maintain chlorine levels: **Free residual chlorine: 1–3 mg/L**
- pH: **7.2 – 7.8**

- Water filtration through sand filters (backwashing every 2–3 days).
- Replace part of pool water daily.

B. Hygiene Measures

- Mandatory shower before entering.
- Footbath with disinfectant.
- Ban spitting, littering, or urination in pool area.

C. Infrastructure

- Smooth, non-slippery floors.
- Separate changing rooms for men/women.
- Lifeguards and safety equipment must be present.

D. Vector Control

- Prevent stagnation of water around pool periphery.

4. Sanitation for Social Gatherings (Melas and Fairs)





17th September - 2nd October
A Swachh Bharat Mission Initiative

Clean Green Festivities in Action!



TELANGANA



A. Water Supply

- Provide **safe drinking water points** at multiple locations.
- Temporary tanks should be chlorinated (0.5 mg/L residual chlorine).

B. Toilets & Mobile Sanitation Units

- Use mobile toilets or temporary latrines.
- Minimum **1 toilet per 100 people**.
- Separate units for men and women.
- Continuous cleaning and disinfection.

C. Waste Management

- Increase number of garbage bins and ensure frequent collection.
- Use **temporary compost pits** for biodegradable waste.

D. Food Safety

- Vendors must follow FSSAI standards.
- Periodic inspection for contamination, cleanliness, expired food items.

E. Crowd & Environmental Control

- Adequate drainage to prevent waterlogging.
- Fogging for mosquito control.

5. Sanitation in Schools and Institutions





A. Drinking Water Safety

- Provide safe drinking water (RO/UV).
- Test water quality quarterly.
- Use covered containers with taps.

B. Toilet Facilities

- As per **Swachh Vidyalaya Guidelines**:
 - 1 toilet for every 40 students.
 - Separate and functional toilets for boys and girls.

C. Personal Hygiene Promotion

- Conduct hygiene education programs.
- Handwashing stations with soap.
- Awareness on menstrual hygiene management for girls.

D. Solid Waste Management

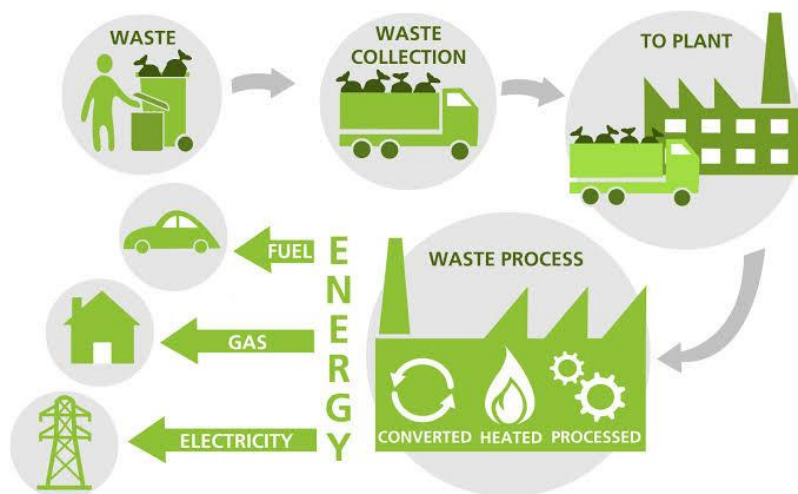
- Introduction of **2-bin/3-bin system**.

- Composting of biodegradable waste.
- Paper recycling programs.

E. Campus Cleanliness

- Daily sweeping of classrooms and playgrounds.
- Control of stray animals.

6. Rural Sanitation – Low-Cost Waste Disposal Methods



1. Pit Latrines

A pit latrine is the simplest and most affordable on-site sanitation method, widely used in rural areas.

Types

1. Simple Pit Latrine
2. Ventilated Improved Pit (VIP) Latrine
3. Twin Pit Pour-Flush Latrine

Components

- Pit: 1–1.5 m diameter, 2–3 m deep
- Slab (stone/concrete)
- Superstructure for privacy
- Optional vent pipe (in VIP latrine)

Working

Human excreta fall directly into the pit. Liquids seep into soil; solids decompose biologically.

Advantages

- Very low cost
- Easy construction
- Suitable for areas with low water availability

Limitations

- Not suitable in high groundwater table
- May cause soil contamination if poorly designed

2. Soak Pit / Leach Pit

Used for **greywater disposal** (bathroom, kitchen wastewater) or for effluent from single-pit latrines.

Structure

- Circular pit (1 m dia, 1–1.5 m deep)
- Filled with bricks, stones, gravel
- Covered with a concrete slab

Working

Wastewater infiltrates the soil through the porous media.

Advantages

- Simple and low cost
- Reduces stagnant wastewater around houses
- Improves environmental sanitation

Limitations

- Requires permeable soil
- Not effective in clayey soils

3. Composting Toilets

These toilets convert human excreta into **safe compost** using aerobic decomposition.

Types

- Single-chamber composting toilet
- Double-chamber composting toilet

Process

- Feces + organic cover material (ash, sawdust, leaves)
- Aerobic decomposition
- After 6–12 months, safe compost is removed

Advantages

- Zero water requirement
- Produces valuable manure
- No pollution to groundwater

Limitations

- Proper management required
- Needs user awareness

4. Biogas Toilets

Integration of toilet waste with a **biogas digester**.

Working

Human excreta + animal dung (optional) → anaerobic digestion → **biogas + slurry**.

Advantages

- Provides free cooking gas
- Sanitary disposal of excreta
- Slurry can be used as fertilizer

Limitations

- Initial cost slightly higher than pit latrines
- Requires regular feeding of digester

5. Ecological Sanitation (EcoSan Toilets)

EcoSan toilets aim to **reuse nutrients** from human waste without water pollution.

Principles

- Urine diversion
- Dehydration of feces
- Safe reuse of sanitized waste in agriculture

Advantages

- Saves water
- Nutrients recovered for soil
- Prevents groundwater contamination

Limitations

- Users need training
- Higher initial cost than simple pit toilets

6. Burial Method / Garbage Pit

Used for **solid waste disposal** in rural houses.

Method

- Dig a pit (1–1.5 m deep)
- Deposit biodegradable waste daily
- Cover with soil periodically
- When full, close the pit and allow decomposition

Advantages

- Very low cost
- Suitable for household-level waste

Limitations

- Not suitable for non-biodegradable waste
- May attract insects if uncovered

7. Community Composting

A low-cost method for villages to manage biodegradable waste collectively.

Steps

1. Collection of organic waste
2. Placement in compost pits/windrows
3. Periodic turning

4. Compost ready in 2–3 months

Advantages

- Reduces waste volume
- Provides compost for farms
- Community involvement

8. Vermicomposting

Uses earthworms to convert organic waste into nutrient-rich compost.

Requirements

- Vermicompost bed or tank
- Moist organic waste
- Red earthworms (*Eisenia foetida*)

Advantages

- Fast composting
- Produces high-quality manure
- Eco-friendly

Limitations

- Requires moderate moisture & shade
- Cannot process plastics or metals

9. Incineration (Small-Scale)

Used for **medical waste, sanitary napkins, and small quantities of dry waste.**

Features

- Low-cost drum incinerators used in rural health centres
- Burns waste at high temperatures

Advantages

- Immediate destruction of waste
- Useful for infection control

Limitations

- Produces smoke
- Not suitable for large-scale waste

10. Drainage and Greywater Management

Proper drainage prevents stagnant water and mosquito breeding.

Low-cost systems

- Kutcha drains (lined with stones)
- Shallow open drains
- Banana plantation soak trenches
- Kitchen garden irrigation using greywater

11. Sanitary Practices in Rural Areas

- Handwashing with soap
- Safe storage of water
- Avoiding open defecation
- Waste segregation at source
- Use of covered bins for refuse
- Community participation & awareness programs

Summary Table

Method	Suitable For	Cost	Advantages	Limitations
Pit Latrine	Excreta	Very Low	Simple, water-saving	Groundwater risk
Soak Pit	Greywater	Low	Easy to build	Needs permeable soil
Compost Toilet	Excreta	Low	Produces compost	Needs management
Biogas Toilet	Excreta + Dung	Medium	Produces gas & manure	Higher initial cost
Garbage Pit	Solid Waste	Very Low	Simple	Only biodegradable waste
Community Composting	Village waste	Low	Compost generation	Requires coordination

One-page exam revision sheet (Rural Sanitation-low cost waste disposal methods)

1. Pit Latrines

Types:

- Simple Pit Latrine
- VIP Latrine (Vent pipe removes odor)
- Twin-Pit Pour-Flush (alternate use)

Key Points:

- 1–1.5 m dia, 2–3 m deep
- Liquids infiltrate; solids decompose
- Low cost, minimal water use

Limitations: High water table, risk of contamination.

2. Soak Pit / Leach Pit

Use: Greywater disposal (bath, kitchen).

Structure: 1 m dia pit filled with stones/bricks.

Advantage: Reduces stagnant wastewater.

Limitation: Needs permeable soil.

3. Composting Toilets

Process: Feces + organic cover → aerobic decomposition → compost in 6–12 months.

Advantages: Zero water, no groundwater pollution, manure recovery.

Note: Needs proper maintenance.

4. Biogas Toilets

Concept: Excreta + dung → anaerobic digestion → biogas + slurry.

Advantages: Free gas, fertilizer, hygienic disposal.

Limitation: Higher initial cost, requires regular feeding.

5. EcoSan Toilets

Principles:

- Urine diversion
- Dehydration of feces
- Safe agricultural reuse

Benefits: Saves water, nutrient recycling, prevents groundwater contamination.

6. Garbage Pit / Burial Method

Use: Household biodegradable waste.

Method: Waste deposited & covered with soil.

Advantages: Extremely low cost, simple.

Limitation: Not for non-biodegradable waste.

7. Community Composting

Steps: Collect → pit/windrow → turn → compost in 2–3 months.

Benefits: Reduces waste load, produces manure, community involvement.

8. Vermicomposting

Key Points: Earthworms convert organic waste to nutrient-rich compost.

Advantages: Fast process, high-quality manure.

Limitations: Needs moisture & shade.

9. Small-Scale Incineration

Use: Sanitary napkins, medical waste.

Advantage: Quick destruction.

Limitation: Smoke emissions.

10. Greywater Drainage Methods

- Kutcha drains
- Soak trenches
- Banana plantation soakage
- Kitchen garden reuse

Prevents stagnation & mosquito breeding.

11. Essential Sanitation Practices

- Avoid open defecation
- Handwashing with soap
- Waste segregation
- Proper water storage
- Use covered dustbins
- Promote community participation

